

# A ONE-DIMENSIONAL WALL MODEL FOR THE $H^s$ EULER–ARNOLD EQUATIONS: BLOWUP BELOW $s = \frac{1}{2}$ , REGULARITY ABOVE

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ABSTRACT. The proved blowup of the three-dimensional Euler equation, due to Chen and Hou, lives on the wall of a cylinder. The preceding paper showed that no symmetry of  $\mathbb{R}^3$  supplies such a wall for the Euler–Arnold family with inertia symbol  $|k|^{2s}$ . Here we put the family on a cylinder. On a domain with a wall the inertia needs boundary conditions. If the swirl component is given Dirichlet conditions, the swirl vanishes on the wall and the Luo–Hou scenario cannot occur. With slip (Neumann) conditions the scenario is available, and restricting to the wall in the manner of Hou and Luo gives a one-dimensional model. On the wall the Jacobian source of the two-and-a-half-dimensional class reduces to one term, which Hou and Luo’s column ansatz removes. What remains is Hou and Luo’s model with its velocity law smoothed by  $2s$  derivatives:  $u_x = c_s H \Lambda^{-2s} \omega$ , with an explicit constant  $c_s$ .

Write  $\alpha = 2s$ . For this model, and for its local version in the manner of Choi, Kiselev and Yao,  $u = -x \int_x^1 \omega(y) y^{\alpha-1} dy$ , we prove the same sharp dichotomy. For  $0 < \alpha < 1$  there is smooth sign-definite data that blows up in finite time. For  $\alpha \geq 1$  every solution with data of that class is global. The threshold  $\alpha = 1$ , that is  $s = \frac{1}{2}$ , is exactly the scaling threshold of the preceding paper away from the axis.

Blowup survives a depletion term, of size  $-\alpha^2 A^2$  in both models, which is absent at  $\alpha = 0$  and which neither the argument of Choi, Kiselev and Yao nor the functional argument of Choi, Hou, Kiselev, Luo, Šverák and Yao controls. We control it by a Riccati comparison along characteristics. For the full law this needs a lower bound on a quadratic form of the vorticity, which we prove from explicit properties of the kernel. We had registered the expectation that the full law could not be pushed to the threshold, and it was wrong.

With the buoyancy the reduction actually produces, the parity of the swirl decides the threshold. It is  $s = \frac{1}{2}$  when the swirl is even about the stagnation point. It is  $s = \frac{1}{4}$  when the swirl is odd, which is the parity of Luo and Hou’s data. In the odd class, blowup below  $s = \frac{1}{4}$  is proved for both laws, and regularity above it for the local law. For the full law with odd swirl, the range  $\frac{1}{4} \leq s < \frac{1}{2}$  remains open. Registered numerical checks, and runs labelled post hoc, agree with the theorems.

## 1. INTRODUCTION

Let  $A$  be the Fourier multiplier with symbol  $|k|^{2s}$ , and consider the Euler–Arnold equation of the right-invariant metric  $\langle Au, u \rangle$  on volume-preserving diffeomorphisms,

$$\partial_t m + (u \cdot \nabla) m + (\nabla u)^T m = -\nabla p, \quad m = Au, \quad \nabla \cdot u = 0. \quad (1)$$

At  $s = 0$  this is the Euler equation. The preceding paper [14] reduced (1) to axisymmetric flows with swirl. It found two scaling thresholds for the conserved sup norm of the swirl circulation:  $s = 1$  on the axis and  $s = \frac{1}{2}$  away from it. It also found that on  $\mathbb{R}^3$  no symmetry produces a wall. This matters because the one proved blowup of the three-dimensional Euler equation from smooth data, due to Chen and Hou [1, 2], sits on the wall of a cylinder. It follows the scenario of Luo and Hou [12]. That scenario is off the axis, so the relevant threshold for it is  $s = \frac{1}{2}$ . As we find in Section 10, this holds for one parity of the swirl.

This paper asks what happens to the wall scenario when the family is placed on a cylinder. We work at the level of the one-dimensional models that preceded the Chen–Hou proof: Hou and Luo’s model [8], and its local version due to Choi, Kiselev and Yao [5]. For both, blowup is proved at  $s = 0$  [5, 6, 4]. The literature contains variants with a different shape of Biot–Savart kernel [10, 7] and with rougher data [3], but, as far as we could find, none whose velocity law has a different order. The Voigt-regularised Boussinesq systems [11, 9] are related, but they are not of Euler–Arnold form, and nothing transfers.

## Results.

- *The wall model* (Section 2). On a cylinder the inertia needs boundary conditions. The stream function is Dirichlet, and we give the swirl Neumann (slip) conditions. With Dirichlet swirl the swirl vanishes on the wall and the Luo–Hou scenario is excluded (Remark 2.1). On the wall, the Jacobian source of the two-and-a-half-dimensional class [13] reduces to  $\partial_y G \partial_x g$  (Lemma 2.2). Under Hou and Luo’s column ansatz this term vanishes, and the wall velocity is  $u_x = c_s H \Lambda^{-2s} \omega$  with  $c_s = \Gamma(s + \frac{1}{2})/(\sqrt{\pi} \Gamma(s + 1))$  (Proposition 2.3). At  $s = 0$  this is Hou and Luo’s model.
- *The local model* (Section 3). For odd vorticity the law has the local part

$$u = -\kappa_\alpha x \int_x^\infty \omega(y) y^{\alpha-1} dy,$$

with  $\alpha = 2s$  and  $\kappa_\alpha \rightarrow 2/\pi$  as  $\alpha \rightarrow 0$  (Lemma 3.1). This gives the model  $\text{CKY}_\alpha$  of (5). At  $\alpha = 0$  it is the model of Choi, Kiselev and Yao. Its scaling threshold is  $\alpha = 1$ .

- *Dichotomy for the local model* (Theorems 5.1 and 6.1). For  $0 < \alpha < 1$ ,  $\text{CKY}_\alpha$  blows up in finite time from suitable smooth sign-definite data. For  $\alpha \geq 1$  it is globally regular for all data of that class.
- *Dichotomy for the full law* (Section 7). The same holds for  $\text{HL}_\alpha$ , with velocity  $u = -c\Lambda^{-1-\alpha}\omega$ , for  $0 < \alpha < 1$  and  $1 \leq \alpha < 2$  respectively (Theorems 7.6 and 7.7). The key is a lower bound  $-c_\alpha A^2$  for a quadratic form of the vorticity (Lemma 7.4, Corollary 7.5).
- *Parity* (Section 8). With the derived buoyancy  $2(B * G)G_x$ , the threshold is  $s = \frac{1}{2}$  for even swirl (Theorem 8.3) and  $s = \frac{1}{4}$  for odd swirl (Theorem 8.4). The odd case is fully proved for the local law. For the full law it is proved except for  $\frac{1}{4} \leq s < \frac{1}{2}$ .
- *Numerics* (Section 9). The predictions were registered before any analysis or run. Two registered checks failed at  $\alpha = 0.5, 0.75$  because of a compression budget fixed in advance. A post hoc run with a larger budget, reported as such, shows blowup there.

What this does not do: the theorems concern one-dimensional models, not (1) on a cylinder. One modelling step remains, the column ansatz of Hou and Luo.

## 2. THE WALL MODEL

We use the reduction of [14]. Let  $\Gamma_m = rm^\theta$  and let  $\Gamma_u = ru^\theta$ . Let  $w^\theta$  be the azimuthal component of  $\nabla \times m$ , and let  $J(f, g) = f_r g_z - f_z g_r$ . For axisymmetric solutions of (1),

$$D_t \Gamma_m = 0, \quad D_t \left( \frac{w^\theta}{r} \right) = \frac{J(\Gamma_m, \Gamma_u)}{r^3} + \frac{2\Gamma_u \partial_z \Gamma_m}{r^4}, \quad (2)$$

with  $D_t = \partial_t + u^r \partial_r + u^z \partial_z$ . The swirl and the poloidal stream function  $\psi$  are recovered as  $\Gamma_u = r(-L_1)^{-s}(\Gamma_m/r)$  and  $\psi = (-L_1)^{-1-s}w^\theta$ , where  $L_1 = \partial_{rr} + r^{-1}\partial_r - r^{-2} + \partial_{zz}$ .

**2.1. Boundary conditions.** On the cylinder  $r < 1$  the fractional powers of  $-L_1$  need boundary conditions. For  $\psi$  the condition is forced: no penetration means  $\psi = 0$  at  $r = 1$ , and we take the Dirichlet spectral power. For the swirl there is a choice.

*Remark 2.1* (Dirichlet swirl excludes the wall scenario). Every function in the range of a Dirichlet spectral power vanishes at  $r = 1$ . With Dirichlet swirl,  $u^\theta = 0$  on the wall for all time. In the Luo-Hou scenario the swirl is largest on the wall, so that choice excludes it. The Fourier-Bessel solver of [14] uses the  $J_1$  Dirichlet basis. It therefore cannot represent this scenario, and it was used there only for the interior one.

We therefore take the Neumann (slip) spectral power for the swirl. Near a point of the wall, freeze  $r = 1$  and write  $x = z$  for the tangential coordinate and  $y = 1 - r$  for the normal one, with the fluid in  $y > 0$ . Write  $G = \Gamma_m$  and  $g = \Gamma_u$ , and let  $w$  denote  $w^\theta/r$ . Then (2) becomes

$$D_t G = 0, \quad D_t w = J(G, g) + 2g \partial_x G, \quad g = (-\Delta_N)^{-s} G, \quad \psi = (-\Delta_D)^{-1-s} w, \quad (3)$$

on the half-plane. Away from the wall, (3) is the system of [13] plus a Boussinesq term.

**Lemma 2.2** (the Jacobian on the wall). *Let  $s > 0$ , and let  $G$  be smooth up to the wall. With Neumann swirl,  $\partial_y g = 0$  on  $y = 0$ , so  $J(G, g) = \partial_y G \partial_x g$  there.*

*Proof.* The even extension of  $g$  is  $\Lambda^{-2s}$  applied to the even extension of  $G$ . That extension is Lipschitz, so its image is  $C^1$  and even in  $y$ , and  $\partial_y g = 0$  at  $y = 0$ . Then  $J = G_y g_x - G_x g_y = G_y g_x$ .  $\square$

The lemma does not make  $J$  vanish. The circulation  $G$  is transported, not produced by a Neumann operator, and  $\partial_y G$  is generated at the wall by the term  $\partial_y u^x \partial_x G$  in the equation for  $\partial_y G$ . The term  $G_y g_x$  vanishes under the column ansatz used below, and only there. Under that ansatz only the buoyancy  $2g \partial_x G$  survives on the wall, so the source of [13] is removed by the same assumption that produces the velocity law. At  $s = 0$ ,  $g = G$ , and  $J$  vanishes identically. Then  $2GG_x = \partial_x(G^2)$ , which is the buoyancy of Luo and Hou with  $\theta = G^2 = u_1^2$ .

**2.2. The velocity law.** Hou and Luo [8] close the system on the wall with a column ansatz: near the wall the vorticity does not depend on the normal variable. We make the same ansatz.

**Proposition 2.3** (wall velocity). *Let  $w(x, y) = \omega(x)$  for  $y > 0$ , and let  $\psi = (-\Delta_D)^{-1-s} w$  on the half-plane. The tangential wall velocity  $u = \partial_y \psi(\cdot, 0)$  satisfies*

$$\hat{u}(\xi) = c_s |\xi|^{-1-2s} \hat{\omega}(\xi), \quad c_s = \frac{\Gamma(s + \frac{1}{2})}{\sqrt{\pi} \Gamma(s + 1)}.$$

*Up to the orientation convention of [8], this means  $u = -c_s \Lambda^{-1-2s} \omega$ , that is  $u_x = c_s H \Lambda^{-2s} \omega$ . Here  $H$  is the Hilbert transform. Also,  $c_0 = 1$ , and a column  $G(x)$  has  $g = \Lambda^{-2s} G$ .*

*Proof.* The Dirichlet spectral power on the half-plane acts on odd extensions in  $y$ . The odd extension of  $w$  is  $\omega(x) \operatorname{sgn} y$ , whose Fourier transform in  $y$  is  $2/(i\eta)$ . Hence

$$\partial_y \hat{\psi}(\xi, 0) = \frac{1}{2\pi} \int_{\mathbb{R}} i\eta \frac{2}{i\eta} \frac{\hat{\omega}(\xi)}{(\xi^2 + \eta^2)^{1+s}} d\eta = \frac{\hat{\omega}(\xi)}{\pi} \int_{\mathbb{R}} \frac{d\eta}{(\xi^2 + \eta^2)^{1+s}} = c_s |\xi|^{-1-2s} \hat{\omega}(\xi),$$

because  $\int_{\mathbb{R}} (1 + t^2)^{-1-s} dt = \sqrt{\pi} \Gamma(s + \frac{1}{2}) / \Gamma(s + 1)$ . The Neumann power acts on even extensions, and the even extension of a column is the column itself.  $\square$

The formula was checked independently in three ways. The first is quadrature at seven values of  $s$ , which agrees to  $10^{-16}$ . The second is the exact solution of  $(k^2 - \partial_y^2)^{1+s} \psi = 1$  at  $s = 0$  and

$s = 1$ , which gives  $\psi'(0) = 1/k$  and  $1/(2k^3)$ , as predicted. The third is a Dirichlet sine series at  $s = \frac{1}{4}$ , which agrees to  $10^{-7}$  (script `derive_wall.py`).

Together, these give the wall model:

$$G_t + uG_x = 0, \quad \omega_t + u\omega_x = 2(\Lambda^{-2s}G)G_x, \quad u_x = c_s H \Lambda^{-2s} \omega. \quad (4)$$

At  $s = 0$  this is exactly Hou and Luo's model. For  $s > 0$  two things change: the velocity law is smoother by  $2s$  derivatives, and the buoyancy carries the factor  $g/G$ . For odd  $G \geq 0$  on the half-period and  $2s < 1$  this factor is positive, because the odd Riesz kernel  $|x-y|^{2s-1} - |x+y|^{2s-1}$  is positive.

### 3. THE LOCAL LAW AND THE MODEL $\text{CKY}_\alpha$

Write  $\alpha = 2s$ . The Riesz potential  $\Lambda^{-b}$  on  $\mathbb{R}$  has kernel  $c'_b|x|^{b-1}$  with  $c'_b = \Gamma(\frac{1-b}{2})/(2^b\sqrt{\pi}\Gamma(\frac{b}{2}))$ , and  $c'_{1+\alpha} < 0$  for  $0 < \alpha < 2$ .

**Lemma 3.1** (local law). *Let  $0 < \alpha < 1$  and let  $\omega$  be odd, smooth and rapidly decaying. Then  $u = -c_s \Lambda^{-1-\alpha} \omega$  is*

$$u(x) = -c_s |c'_{1+\alpha}| \int_0^\infty \omega(y) (|x+y|^\alpha - |x-y|^\alpha) dy,$$

and as  $x \rightarrow 0$ ,

$$\frac{u(x)}{x} \rightarrow -\kappa_\alpha \int_0^\infty \omega(y) y^{\alpha-1} dy, \quad \kappa_\alpha = 2\alpha c_s |c'_{1+\alpha}| \rightarrow \frac{2}{\pi} (\alpha \rightarrow 0).$$

*Proof.* The first formula is the kernel representation, symmetrised using oddness. For  $y \gg x$ ,  $|x+y|^\alpha - |x-y|^\alpha = 2\alpha xy^{\alpha-1} + O(x^3 y^{\alpha-3})$ . The limit of  $\kappa_\alpha$  follows from  $\Gamma(-\alpha/2) \sim -2/\alpha$ .  $\square$

A fast Fourier transform check against the kernel agrees to between  $5 \times 10^{-6}$  and  $3 \times 10^{-4}$ , with the residual scaling like  $L^{\alpha-1}$  in the box size  $L$ . The limit is reproduced to six digits at  $x = 10^{-3}$ , and  $\kappa_\alpha$  tends to  $2/\pi$  (`derive_wall.py`). In the manner of [5] we keep the main term and scale out the constant. On  $[0, 1]$  this gives

$$\begin{aligned} \rho_t + u\rho_x &= 0, & \omega_t + u\omega_x &= \rho_x, \\ u(t, x) &= -x\Omega(t, x), & \Omega(t, x) &= \int_x^1 \omega(t, y) y^{\alpha-1} dy. \end{aligned} \quad (5)$$

We call this  $\text{CKY}_\alpha$ . At  $\alpha = 0$  it is the model of [5]. Following [5] it uses the buoyancy  $\rho_x$  of [8], not the factor  $g/G$  of (4). We write  $\eta = \rho_x$ , which satisfies  $\eta_t + (u\eta)_x = 0$ .

**Scaling.** Under  $x \mapsto \lambda x$  and  $t \mapsto \mu t$  with  $\|\rho\|_{L^\infty}$  fixed, we have  $u \sim \lambda/\mu$  and  $\omega \sim \lambda^{-\alpha}/\mu$ , and the balance  $\omega_t \sim \rho_x$  forces  $\mu^2 \sim \lambda^{1-\alpha}$ . Concentration takes vanishing time exactly when  $\alpha < 1$ . This is the off-axis count of [14] with  $\alpha = 2s$ .

### 4. STRUCTURE OF $\text{CKY}_\alpha$

**Definition 4.1.** The data class  $\mathcal{D}$  consists of pairs  $(\omega_0, \eta_0)$  of  $C^\infty$  functions compactly supported in  $(0, 1)$  with  $\omega_0 \geq 0$  and  $\eta_0 \geq 0$ . We set  $\rho_0(x) = \int_0^x \eta_0$ .

**Lagrangian form and local theory.** Let  $\phi(t, X)$  be the flow of  $u$  and let  $j = \partial_X \phi$ . Along labels  $X$  in a compact interval  $[a, b] \subset (0, 1)$  containing the supports,

$$\dot{\phi} = -\phi \Omega, \quad \dot{j} = (-\Omega + \omega \phi^\alpha) j, \quad \dot{\omega} = \frac{\eta_0(X)}{j}, \quad \Omega(t, \phi(t, X)) = \int_X^1 \omega \phi^{\alpha-1} j dX'. \quad (6)$$

The first two equations use  $u_x = -\Omega - x\Omega_x = -\Omega + x^\alpha \omega$ . For labels  $X < a$  we have  $\Omega = \Omega(t, 0)$ , and for  $X > b$  we have  $\Omega = 0$ , so the flow is explicit there.

**Proposition 4.2** (local theory). *Let  $k \geq 1$  and let the data be in  $\mathcal{D}$ , with supports in  $[a, b] \subset (0, 1)$ . There is a unique solution of (6) in  $C^1([0, T]; C^k([a, b])^3)$  on a maximal interval  $[0, T_{\max})$ . The corresponding Eulerian fields  $\omega(t, \cdot)$  and  $\eta(t, \cdot)$  are  $C^k$  with support in  $[\phi(t, a), \phi(t, b)]$ . If  $T_{\max} < \infty$ , then one of the quantities*

$$\left(\inf_X \phi\right)^{-1}, \quad \sup_X (j + j^{-1}), \quad \sup_X |\omega| \quad (7)$$

*is unbounded on  $[0, T_{\max})$ . Equivalently, the solution continues as long as  $\int_0^T \|u_x\|_\infty dt < \infty$ .*

*Proof.* The right-hand side of (6) maps the open set  $\mathcal{U} = \{\phi > 0, j > 0\} \subset C^k([a, b])^3$  to  $C^k([a, b])^3$ , and it is locally Lipschitz there. The nonlocal term is an integral in  $X$ , which gains a derivative, and  $\phi^{\alpha-1}$  and  $\phi^\alpha$  are smooth on  $\phi > 0$ . Picard's theorem in the Banach space  $C^k([a, b])^3$  gives local existence and uniqueness. The solution continues as long as it stays in a closed bounded subset of  $\mathcal{U}$ .

At order zero this is the boundedness of (7), because  $0 \leq \Omega \leq (\inf \phi)^{\alpha-1} \int |\omega| j dX$  and  $\phi \leq 1$ . At order  $m \leq k$ , differentiate (6)  $m$  times in  $X$ , using  $\partial_X \Omega = -\omega \phi^{\alpha-1} j$ . The result is a linear system for  $(\partial_X^m \phi, \partial_X^m j, \partial_X^m \omega)$  whose coefficients and forcing involve only derivatives of order below  $m$ . By induction and Gronwall, all  $C^k$  norms stay bounded on bounded intervals once (7) does.

The map  $X \mapsto \phi(t, X)$  is a  $C^k$  diffeomorphism because  $j > 0$ , so the Eulerian fields are  $C^k$ . Finally,  $j = \exp \int_0^t u_x$ ,  $\omega$  grows at most by  $\int \|\eta\|_\infty$  with  $\eta = \eta_0/j$ , and  $\phi(t, a) \geq a \exp(-\int_0^t \|u_x\|_\infty)$ . So  $\int_0^T \|u_x\|_\infty < \infty$  bounds (7). Conversely,  $\|u_x\|_\infty \leq \Omega(t, 0) + \sup \omega \phi^\alpha$  is bounded by (7).  $\square$

A classical solution on  $[0, T]$  has a Lipschitz velocity with  $u(t, 0) = 0$ , so  $\phi(t, X) \geq X \exp(-\int_0^t \|u_x\|_\infty) > 0$ .

**Signs.** For data in  $\mathcal{D}$  the density  $\eta = \eta_0/j$  stays nonnegative. Then  $\omega(t, \phi) = \omega_0 + \int_0^t \eta \geq 0$ , so  $\Omega \geq 0$ ,  $u \leq 0$ , and every characteristic moves toward the origin.

**Lemma 4.3** (the identity). *Along any characteristic  $\phi(t) = \phi(t, X)$ , including  $\phi \equiv 0$ ,*

$$\frac{d}{dt} \Omega(t, \phi(t)) = \int_\phi^1 \omega^2 y^{2\alpha-1} dy - \frac{\alpha}{2} \Omega(t, \phi)^2 + \int_\phi^1 \eta y^{\alpha-1} dy. \quad (8)$$

*Proof.* We have  $\frac{d}{dt} \Omega(t, \phi) = \int_\phi^1 \omega_t y^{\alpha-1} dy + \omega(\phi) \phi^\alpha \Omega(\phi)$ , and  $\omega_t = -u \omega_y + \eta = y \Omega \omega_y + \eta$ . Integrating by parts, using  $\Omega_y = -\omega y^{\alpha-1}$  and  $\omega(t, 1) = \Omega(t, 1) = 0$ ,

$$\int_\phi^1 \Omega \omega_y y^\alpha dy = -\Omega \omega \phi^\alpha + \int_\phi^1 \omega^2 y^{2\alpha-1} dy - \alpha \int_\phi^1 \omega \Omega y^{\alpha-1} dy.$$

The last integral is  $-\int_\phi^1 \Omega \Omega_y dy = \frac{1}{2} \Omega(\phi)^2$ . The boundary term cancels.  $\square$

At  $\alpha = 0$ , (8) is Lemma 4.1 of [5]. For  $\alpha > 0$  the term  $-\frac{\alpha}{2}\Omega^2$  is new. The argument of [5] uses the lower bound  $\frac{d}{dt}\Omega \geq -4$  and cannot absorb it. The positive term  $\int \omega^2 y^{2\alpha-1}$  does not help: for profiles  $\omega \sim y^{-\alpha}$  spread over many scales, the negative term dominates it.

## 5. BLOWUP FOR $0 < \alpha < 1$

**Theorem 5.1.** *Let  $0 < \alpha < 1$ . There is data in  $\mathcal{D}$  whose solution of  $CKY_\alpha$  cannot be continued as a classical solution past a finite time  $T_* \leq \frac{3}{2}$ . That is,  $T_{\max} \leq \frac{3}{2}$  in Proposition 4.2, and  $\int_0^{T_{\max}} \|u_x\|_\infty dt = \infty$ .*

*Proof. Data.* Take  $\eta_0 \geq 0$  supported in  $[\frac{1}{4}, \frac{1}{2}]$  and positive inside, with  $\int \eta_0 = 2$ . Then  $\rho_0$  increases strictly from 0 to 2 on  $[\frac{1}{4}, \frac{1}{2}]$ . For  $n \geq 1$  let  $x_n \in (\frac{1}{4}, \frac{1}{2})$  be defined by  $\rho_0(x_n) = \rho_n := \frac{1}{2} + 2^{-n}$ . Then  $x_1 > x_2 > \dots \rightarrow x_\infty > \frac{1}{4}$  and  $\rho_0(x_\infty) = \frac{1}{2}$ . Take  $\omega_0 = M\chi$ , with  $\chi \geq 0$  smooth, not identically zero, and supported in  $(\frac{1}{2}, 1)$ . The constant  $M$  is fixed below.

*Notation.* Let  $\Phi_n(t) = \phi(t, x_n)$ ,  $\psi_n = -\ln \Phi_n$ , and  $\Omega_n(t) = \Omega(t, \Phi_n(t))$ . Then  $\dot{\psi}_n = \Omega_n \geq 0$ , so each  $\psi_n$  is nondecreasing, and  $\psi_n \geq \psi_{n-1}$  because characteristics keep their order.

*The inequality.* In (8) along  $\Phi_n$ , drop the first term and keep only the part of the last term over  $[\Phi_n, \Phi_{n-1}]$ . There  $\rho$  increases from  $\rho_n$  to  $\rho_{n-1}$ , and  $y^{\alpha-1} \geq \Phi_{n-1}^{\alpha-1}$  because  $\alpha < 1$ . So for  $n \geq 2$ ,

$$\dot{\Omega}_n \geq 2^{-n} e^{(1-\alpha)\psi_{n-1}} - \frac{\alpha}{2} \Omega_n^2. \quad (9)$$

*The Riccati step.* Let  $\tau_n = 2^{-n}$ ,  $t_1 = 1$ ,  $t_n = t_{n-1} + \tau_n$ , and  $T_* = \lim t_n = \frac{3}{2}$ . Set  $a_n = \psi_n(t_n)$ . On  $[t_{n-1}, t_n]$  we have  $\psi_{n-1} \geq a_{n-1}$ , so (9) gives  $\dot{\Omega}_n \geq F_n - \frac{\alpha}{2} \Omega_n^2$  with  $F_n = 2^{-n} e^{(1-\alpha)a_{n-1}}$  and  $\Omega_n(t_{n-1}) \geq 0$ . The solution of  $\dot{v} = F - \frac{\alpha}{2} v^2$  with  $v(0) = 0$  is  $v(\sigma) = \sqrt{2F/\alpha} \tanh(\sqrt{\alpha F/2} \sigma)$ . By comparison,

$$\psi_n(t_n) - \psi_n(t_{n-1}) \geq \int_0^{\tau_n} v = \frac{2}{\alpha} \ln \cosh\left(\sqrt{\frac{\alpha F_n}{2}} \tau_n\right) \geq \sqrt{\frac{2F_n}{\alpha}} \tau_n - \frac{2 \ln 2}{\alpha}.$$

Since  $\psi_n(t_{n-1}) \geq \psi_{n-1}(t_{n-1}) = a_{n-1}$ ,

$$a_n \geq a_{n-1} + c 2^{-3n/2} e^{b a_{n-1}} - d, \quad b = \frac{1-\alpha}{2}, \quad c = \sqrt{2/\alpha}, \quad d = \frac{2 \ln 2}{\alpha}. \quad (10)$$

*Induction.* Choose  $K$  with  $bK \geq \frac{3}{2} \ln 2$  and  $\frac{c}{8} e^{bK} \geq K + d$ . If  $a_{n-1} \geq K(n-1)$ , then  $2^{-3n/2} e^{b a_{n-1}} \geq 2^{-3/2} (e^{bK} 2^{-3/2})^{n-1} \geq e^{bK}/8$ , and (10) gives  $a_n \geq Kn$ . For the base case  $a_1 \geq K$ , drop the buoyancy in (8). Then  $\dot{\Omega}_1 \geq -\frac{\alpha}{2} \Omega_1^2$ , so  $\psi_1(1) \geq \frac{2}{\alpha} \ln(1 + \frac{\alpha}{2} \Omega_1(0))$ , and  $\Omega_1(0) = M \int \chi y^{\alpha-1}$  can be made as large as needed by the choice of  $M$ .

*Conclusion.* All of the above holds on  $[0, \min(t, T_{\max}))$ . If  $T_{\max} \leq T_*$  there is nothing more to prove. Otherwise the solution is classical on  $[0, T_*]$ . Since  $x_\infty < x_n$ , we get  $-\ln \phi(t, x_\infty) \geq \psi_n(t) \geq a_n \geq Kn$  for  $t \geq t_n$  and every  $n$ . Hence  $\phi(T_*, x_\infty) = 0$ . This contradicts  $\phi(t, x_\infty) \geq x_\infty e^{-\int \|u_x\|} > 0$ . The last claim of the theorem is the continuation criterion of Proposition 4.2.  $\square$

*Remark 5.2.* At  $\alpha = 0$  the depletion term is absent, and the theorem is that of [5] (for their data). Two features of (10) deserve comment. The growth rate  $e^{(1-\alpha)a/2}$  vanishes exactly at the scaling threshold. The Riccati damping caps  $\dot{\psi}_n$  at  $\sqrt{2F_n/\alpha}$ , but it cannot stop the recursion, because  $\int_0^\infty e^{-(1-\alpha)\psi/2} d\psi < \infty$ . As  $\alpha \rightarrow 1$ , the constant  $K$  grows like  $(1-\alpha)^{-1} \log \frac{1}{1-\alpha}$ .

6. REGULARITY FOR  $\alpha \geq 1$ 

**Theorem 6.1.** *Let  $\alpha \geq 1$ . For every datum in  $\mathcal{D}$  the solution of  $CKY_\alpha$  is global and smooth.*

*Proof.* All quantities below are evaluated along a characteristic  $\phi(t) = \phi(t, X)$ .

*Step 1:*  $\eta x$  does not increase. Since  $\dot{\eta} = -u_x \eta$  and  $\dot{\phi} = -\phi \Omega$ ,

$$\frac{d}{dt}(\eta \phi) = \eta \phi (-u_x - \Omega) = -(\eta \phi)(\omega \phi^\alpha) \leq 0.$$

Hence  $\eta(t, x) x \leq M_0 := \sup_x \eta_0(x) x$  for all  $t$ . This step holds for every  $\alpha$ .

*Step 2:*  $\omega x^\alpha$  grows at most linearly. We have

$$\frac{d}{dt}(\omega \phi^\alpha) = \eta \phi^\alpha - \alpha \omega \phi^\alpha \Omega \leq (\eta \phi) \phi^{\alpha-1} \leq M_0,$$

using  $\phi \leq 1$  and  $\alpha \geq 1$ . So  $P(t) := \sup_x \omega x^\alpha \leq P(0) + M_0 t$ .

*Step 3:*  $W = \Omega(t, 0)$  is bounded. Apply (8) with  $\phi \equiv 0$ . The first term satisfies  $\int_0^1 \omega^2 y^{2\alpha-1} = \int_0^1 (\omega y^\alpha)(\omega y^{\alpha-1}) \leq PW$ . The last satisfies  $\int_0^1 \eta y^{\alpha-1} \leq \int \eta = \rho_{\max}$ , using  $\alpha \geq 1$ . So

$$\dot{W} \leq P(t) W - \frac{\alpha}{2} W^2 + \rho_{\max},$$

and  $W(t) \leq \max(W(0), \sup_{\sigma \leq t} \alpha^{-1}(P + \sqrt{P^2 + 2\alpha\rho_{\max}})(\sigma))$ , which is finite for every  $t$ .

*Step 4: continuation.* Since  $\omega \geq 0$ ,  $0 \leq \Omega \leq W$  and  $\|u_x\|_\infty \leq W + P$ . So  $j$  and  $j^{-1}$  are bounded by  $\exp \int (W + P)$ . The inner edge of the support satisfies  $\phi(t, a) \geq a e^{-\int W}$ , and  $\omega \leq P/\phi(t, a)^\alpha$ . Every quantity in (7) therefore stays finite on bounded intervals, and the solution is global.  $\square$

*Remark 6.2.* Step 2 is where  $\alpha \geq 1$  enters. For  $\alpha < 1$  the bound becomes  $M_0 \phi^{\alpha-1}$ , which is unbounded as characteristics approach the origin. This is the growth that drives Theorem 5.1. Step 3 uses the depletion term that the blowup proof had to overcome. The two theorems together give a dichotomy within the sign-definite class: blowup for some data if and only if  $\alpha < 1$ , that is  $s < \frac{1}{2}$ .

## 7. THE FULL LAW

We now drop the second reduction and work with the full law of Proposition 2.3. On the real line, with  $\omega$  odd and  $\theta$  even, consider

$$\theta_t + u \theta_x = 0, \quad \omega_t + u \omega_x = \theta_x, \quad u = -c \Lambda^{-1-\alpha} \omega, \quad (11)$$

which we call  $HL_\alpha$ . By Lemma 3.1, after the constant is absorbed into time, for  $x > 0$

$$u(x) = -x V(x), \quad V(x) = \int_0^\infty M_\alpha\left(\frac{x}{z}\right) d\mu(z), \quad d\mu(z) = \omega(z) z^{\alpha-1} dz, \quad (12)$$

with

$$M_\alpha(s) = \frac{g(s)}{s}, \quad g(s) = (1+s)^\alpha - |1-s|^\alpha.$$

Replacing  $M_\alpha$  by  $2\alpha\chi_{(0,1)}$  gives  $CKY_\alpha$  with velocity  $-2\alpha x A$ , where

$$A(t, x) = \int_x^\infty \omega(t, y) y^{\alpha-1} dy = \mu((x, \infty)).$$

As  $\alpha \rightarrow 0$ ,  $M_\alpha/\alpha$  tends to the kernel  $s^{-1} \log \left| \frac{1+s}{1-s} \right|$  of Hou and Luo's model in the variables of [6, Section 3].

**Definition 7.1.** The class  $\mathcal{D}_{\text{HL}}$  consists of data with  $\omega_0$  odd,  $\theta_0$  even, and  $\omega_0$  and  $\theta'_0$  smooth and compactly supported, such that  $\omega_0 \geq 0$  and  $\theta'_0 \geq 0$  on  $(0, \infty)$ .

As before, the signs propagate. Oddness gives  $u(t, 0) = 0$ , so characteristics do not cross the origin. The density  $\eta = \theta_x$  solves a continuity equation, and  $\omega$  increases along characteristics. Hence  $\mu \geq 0$  and  $V \geq 0$  on  $(0, \infty)$ .

**Proposition 7.2** (local theory for the full law). *Let  $0 < \alpha < 2$  and let the data be in  $\mathcal{D}_{\text{HL}}$ . There is a unique solution with  $\omega(t)$  and  $\theta_x(t)$  in  $C^k$ , compactly supported in  $\mathbb{R} \setminus \{0\}$  away from the origin, on a maximal interval  $[0, T_{\max})$ . It continues as long as  $\int_0^T \|u_x\|_{\infty} dt < \infty$ .*

*Proof.* Work in  $\xi = -\ln x$  on the positive half-line. Write  $\Omega(\xi) = \omega(e^{-\xi})$ ,  $U = -u/x$ , and let  $\rho$  be the density  $\theta_x x$  in  $\xi$ . Then

$$\rho_t + (U\rho)_{\xi} = 0, \quad \Omega_t + U\Omega_{\xi} = e^{\xi}\rho, \quad U = K * (e^{-\alpha\xi}\Omega), \quad K(\zeta) = M_{\alpha}(e^{-\zeta}).$$

Here  $K$  is bounded, and  $K'(\zeta) = -sM'_{\alpha}(s)$  with  $s = e^{-\zeta}$  lies in  $L^1(d\zeta)$ , since  $\int_0^{\infty} |M'_{\alpha}| ds < \infty$ . While the supports stay in a bounded  $\xi$ -interval, the weights  $e^{\xi}$  and  $e^{-\alpha\xi}$  are smooth and bounded.

- *Velocity bounds.*  $\|\partial_{\xi}^j U\|_{\infty} \leq \|K'\|_{L^1} \|\partial_{\xi}^{j-1}(e^{-\alpha\xi}\Omega)\|_{\infty}$  for  $j \geq 1$ .
- *Energy estimates.* Differentiating  $m$  times, the top-order terms  $U\partial_{\xi}^{m+1}\Omega$  and  $U\partial_{\xi}^{m+1}\rho$  are transport terms. All other terms carry at most  $m$  derivatives of  $(\Omega, \rho)$ , with coefficients bounded by  $\|U_{\xi}\|_{\infty}$  and lower norms.

The usual iteration in  $C^k$  therefore gives local existence and uniqueness. Gronwall bounds the  $C^k$  norms on any interval where  $\int \|U_{\xi}\|_{\infty}$  is finite and the supports stay bounded. Both hold if  $\int \|u_x\|_{\infty} < \infty$ : we have  $U_{\xi} = u_x - u/x$ , and  $|u/x| \leq \|u_x\|_{\infty}$  because  $u(0) = 0$ . The inner edge of the support satisfies  $x \geq x_0 e^{-\int \|u_x\|_{\infty}}$ .  $\square$

**Lemma 7.3** (the identity for any law  $u = -xV$ ). *Let  $u = -xV$  with  $V$  smooth on  $(0, \infty)$ . Along any characteristic  $\phi(t)$  in  $(0, \infty)$ ,*

$$\frac{d}{dt}A(t, \phi) = - \int_{\phi}^{\infty} (\alpha V + yV_y) \omega y^{\alpha-1} dy + \int_{\phi}^{\infty} \theta_y y^{\alpha-1} dy.$$

For  $CKY_{\alpha}$ , where  $V = A$ , this is (8).

*Proof.* We have  $\frac{d}{dt}A(t, \phi) = \int_{\phi}^{\infty} \omega_t y^{\alpha-1} dy + \omega(\phi)\phi^{\alpha}V(\phi)$ , and  $\omega_t = yV\omega_y + \theta_y$ . Integrating by parts,  $\int_{\phi}^{\infty} y^{\alpha}V\omega_y dy = -\phi^{\alpha}V(\phi)\omega(\phi) - \int_{\phi}^{\infty} \omega(\alpha y^{\alpha-1}V + y^{\alpha}V_y) dy$ .  $\square$

For  $\text{HL}_{\alpha}$ ,  $\alpha V + xV_x = \int_0^{\infty} P(x/z) d\mu(z)$ , where

$$P(s) = \alpha M_{\alpha}(s) + sM'_{\alpha}(s) = \frac{(\alpha-1)g(s)}{s} + g'(s), \quad P_{\text{sym}}(s) = \frac{1}{2}(P(s) + P(1/s)).$$

**Lemma 7.4** (the kernel). (a) *Let  $0 < \alpha < 1$ . Then  $M_{\alpha} \geq 2\alpha$  on  $(0, 1)$  and  $M_{\alpha} > 0$  on  $(1, \infty)$ .*

(b) *Let  $0 < \alpha < 1$ . Then  $P < 0$  on  $(1, \infty)$ .*

(c) *For  $0 < s < 1$ ,*

$$P(s) + P(1/s) = g'(s)(1 - s^{2-\alpha}) + g(s)\left(\frac{\alpha-1}{s} + (2\alpha-1)s^{1-\alpha}\right).$$

*Consequently, for  $0 < \alpha < 1$ ,  $c_{\alpha} := \sup P_{\text{sym}} \leq \frac{\alpha(3-\alpha)}{2} + 2^{\alpha-1} \max(0, 2\alpha-1)$ .*

(d) *Let  $1 \leq \alpha < 2$ . Then  $P_{\text{sym}} > 0$ , and  $M_{\alpha}$  and  $sM'_{\alpha}$  are bounded.*

*Proof.* (a) The function  $f(t) = (1+t)^\alpha - (1-t)^\alpha$  has  $f'(t) = \alpha((1+t)^{\alpha-1} + (1-t)^{\alpha-1}) \geq 2\alpha$ , because  $r \mapsto r^{\alpha-1}$  is convex. So  $g(s) \geq 2\alpha s$  on  $(0, 1)$ . On  $(1, \infty)$ ,  $(s+1)^\alpha > (s-1)^\alpha$ .

(b) On  $(1, \infty)$  we have  $g'(s) = \alpha((s+1)^{\alpha-1} - (s-1)^{\alpha-1}) < 0$  and  $(\alpha-1)g/s < 0$ .

(c) Since  $g(1/s) = s^{-\alpha}g(s)$ , we have  $M_\alpha(1/s) = s^{1-\alpha}g(s)$  and  $s^{-1}M'_\alpha(1/s) = -s\frac{d}{ds}M_\alpha(1/s)$ . This gives  $P(1/s) = (2\alpha-1)s^{1-\alpha}g - s^{2-\alpha}g'$ , and the formula follows. For  $0 < \alpha < 1$  and  $0 < s < 1$ , we use three bounds:

- $1 - s^{2-\alpha} = (2-\alpha) \int_s^1 t^{1-\alpha} dt \leq (2-\alpha)(1-s)$ ;
- $g' \leq \alpha(1 + (1-s)^{\alpha-1})$ ;
- $g \leq 2^\alpha$ .

The first term of the formula is therefore at most  $\alpha(3-\alpha)$ . The second is at most  $2^\alpha \max(0, 2\alpha-1)$ , because  $(\alpha-1)/s < 0$ . The function  $P_{\text{sym}}$  is invariant under  $s \mapsto 1/s$  and continuous at  $s = 1$ , so the bound holds on  $(0, \infty)$ .

(d) For  $\alpha \geq 1$  and  $0 < s < 1$  every factor in (c) is positive, and  $P_{\text{sym}}$  is invariant under  $s \mapsto 1/s$ . The function  $M_\alpha$  is continuous, tends to  $2\alpha$  at 0, and behaves like  $2\alpha s^{\alpha-2}$  at  $\infty$ . The function  $sM'_\alpha = g' - g/s$  is bounded on  $(0, 1)$  and on  $(1, \infty)$ , and tends to 0 at both ends.  $\square$

Numerically,  $\sup P_{\text{sym}} = \alpha^2$  to all digits computed, for  $\alpha \in \{0.05, \dots, 0.99\}$ . The supremum is approached as  $s \rightarrow 0$  and  $s \rightarrow \infty$  (`kernel_h1.py`). This is exactly the depletion constant of  $\text{CKY}_\alpha$  with velocity  $-2\alpha xA$ . The proofs only use  $c_\alpha < \infty$ .

**Corollary 7.5.** *Let  $0 < \alpha < 1$  and let the data be in  $\mathcal{D}_{\text{HL}}$ . Along every characteristic in  $(0, \infty)$ ,*

$$V(\phi) \geq 2\alpha A(\phi), \quad \frac{d}{dt}A(t, \phi) \geq -c_\alpha A(t, \phi)^2 + \int_\phi^\infty \theta_y y^{\alpha-1} dy.$$

*Proof.* The first inequality is (a). In Lemma 7.3, split the  $z$ -integral at  $\phi$ . The part with  $z \leq \phi < y$  has  $y/z > 1$ , and it is nonnegative by (b). The part with  $y, z > \phi$  can be symmetrised, and by (c) it is at least  $-c_\alpha \mu((\phi, \infty))^2$ . The singularity of  $P$  at  $s = 1$  is like  $|1-s|^{\alpha-1}$  and is integrable.  $\square$

**Theorem 7.6.** *Let  $0 < \alpha < 1$ . There is data in  $\mathcal{D}_{\text{HL}}$  whose solution of  $HL_\alpha$  cannot be continued as a classical solution past a finite time  $T_* \leq \frac{3}{2}$ . Then  $\int_0^{T_{\max}} \|u_x\|_\infty dt = \infty$ , by Proposition 7.2.*

*Proof.* Take  $\theta'_0 \geq 0$  supported in  $[\frac{1}{4}, \frac{1}{2}]$  on the positive side and positive inside, with  $\int_0^\infty \theta'_0 = 2$  and  $\theta_0(0) = 0$ . Take  $\omega_0 = M\chi$  on  $(\frac{1}{2}, 1)$ , extended oddly. Define  $x_n, \Phi_n$  and  $\psi_n$  as in the proof of Theorem 5.1, and set  $A_n = A(t, \Phi_n)$ . Corollary 7.5 gives

$$\dot{\psi}_n = V(\Phi_n) \geq 2\alpha A_n, \quad \dot{A}_n \geq 2^{-n} e^{(1-\alpha)\psi_{n-1}} - c_\alpha A_n^2,$$

because  $\theta_y \geq 0$  on  $(0, \infty)$  and  $y^{\alpha-1} \geq \Phi_{n-1}^{\alpha-1}$  on  $[\Phi_n, \Phi_{n-1}]$ . The Riccati comparison of Theorem 5.1 now gives

$$a_n \geq a_{n-1} + \frac{2\alpha}{\sqrt{c_\alpha}} 2^{-3n/2} e^{(1-\alpha)a_{n-1}/2} - \frac{2\alpha \ln 2}{c_\alpha},$$

which has the form (10). The induction, the base case (using  $\dot{A}_1 \geq -c_\alpha A_1^2$  and a large  $M$ ) and the conclusion are unchanged.  $\square$

**Theorem 7.7.** *Let  $1 \leq \alpha < 2$ . For every datum in  $\mathcal{D}_{\text{HL}}$  the solution of  $HL_\alpha$  is global and smooth.*

*Proof.* On  $(0, \infty)$  we have  $u \leq 0$ , so the supports stay in  $[0, R_0]$ . Apply Lemma 7.3 at the fixed characteristic  $\phi \equiv 0$ . By (d) the quadratic term is  $-\iint P_{\text{sym}} d\mu d\mu \leq 0$ , so

$$\frac{d}{dt}A(t, 0) \leq \int_0^\infty \theta_y y^{\alpha-1} dy \leq R_0^{\alpha-1}(\theta(\infty) - \theta(0)),$$

and  $A(t, 0)$  grows at most linearly. We have  $u_x = -V - xV_x$ ,  $|V| \leq \sup M_\alpha A(t, 0)$  and  $|xV_x| \leq \sup |sM'_\alpha| A(t, 0)$ , so  $\int_0^T \|u_x\|_\infty < \infty$  for every  $T$ . By Proposition 7.2 the solution is global.  $\square$

*Remark 7.8.* For the full law the regularity proof is simpler than for  $\text{CKY}_\alpha$ . The local model's kernel  $2\alpha\chi_{(0,1)}$  has a jump, which puts a negative delta into  $P_{\text{sym}}$  and a positive term  $\int \omega^2 y^{2\alpha-1}$  into (8). The smooth kernel has neither. At  $\alpha = 0$ , normalising by  $1/\alpha$ , we get  $c_0 = 0$ , and Corollary 7.5 is Lemma 3.4 of [6] in these variables. The argument of Section 5 then gives another route to blowup of Hou and Luo's model on the line. The functional argument of [6, Section 3.4] does not close for  $\alpha > 0$ . There the depletion enters as an integral of  $A^2$  against  $\theta_x$ , and Jensen's inequality bounds it in the wrong direction.

## 8. THE DERIVED BUOYANCY, IN BOTH PARITIES

We now replace the buoyancy  $\theta_x$  by the one the reduction actually produces. The model is

$$G_t + uG_x = 0, \quad \omega_t + u\omega_x = 2gG_x, \quad g = B * G, \quad (13)$$

on the line, with  $\omega$  odd. The velocity law is either the local law  $u = -x \int_x^\infty \omega y^{\alpha-1} dy$  of  $\text{CKY}_\alpha$ , or the full law (12). The swirl  $G$  is transported, and  $G_x$  solves the continuity equation. The swirl velocity  $g$  is obtained from  $G$  by a kernel  $B$ . The frozen half-plane gives the Riesz kernel  $c|r|^{2s-1}$ .

For even  $G$  that is bounded but not decaying, the Riesz kernel makes  $B * G$  diverge when  $2s < 1$ . The frozen model has lost the infrared scale of the cylinder, where the Neumann swirl operator has a spectral gap. We therefore allow any kernel of the following class, of which the Riesz kernel (for odd  $G$ ) and the Bessel potential of  $(1 - \partial_x^2)^{-s}$  are examples.

**Definition 8.1.** A kernel  $B$  is *admissible* if it is even, locally integrable and completely monotone on  $(0, \infty)$ . In particular it is then positive, nonincreasing and convex there. For  $s < \frac{1}{2}$  we also ask that  $B(r) \leq b_2 r^{2s-1}$  and  $|B'(r)| \geq b_3 r^{2s-2}$  for  $0 < r \leq r_0$ . For even  $G$  we ask that  $\int_{\mathbb{R}} B = 1$ .

The Bessel potential kernel is  $c r^{(2s-1)/2} K_{(1-2s)/2}(r)$ , which is a product of completely monotone functions. Another example is  $B_e(r) = |r|^{2s-1} e^{-|r|} / (2\Gamma(2s))$ , which we use in the numerics.

**Lemma 8.2** (parity). *Let  $B$  be admissible.*

- (i) *If  $G$  is even with  $G_0 \leq G \leq G_1$  and  $\int B = 1$ , then  $G_0 \leq g \leq G_1$ .*
- (ii) *Suppose  $G$  is odd, nonnegative and nondecreasing on  $(0, \infty)$ , with  $\sup G = G_1$ . Then for  $x > 0$ ,*

$$0 \leq g(x) \leq 2G_1 \int_0^x B.$$

*If  $s < \frac{1}{2}$  and  $0 < x \leq x_0 := \frac{r_0}{3} 2^{-1/(1-2s)}$ , then*

$$g(x) \geq c_s x^{2s} G(x), \quad c_s = \frac{b_3 (3/2)^{2s-2} 2^{2s-1}}{1-2s}.$$

*Proof.* (i)  $g$  is an average of  $G$ .

(ii) We have  $g(x) = \int_0^\infty (B(x-y) - B(x+y)) G(y) dy$ , and the bracket is nonnegative because  $|x-y| \leq x+y$ . With  $G \leq G_1$ , the integral of the bracket is  $\int_{-x}^x B$ . For the lower bound, keep only  $y \in [2x, \frac{2}{3}r_0]$ . There  $B(y-x) - B(y+x) = \int_{y-x}^{y+x} |B'| \geq 2x|B'(y+x)| \geq 2xb_3(\frac{3}{2}y)^{2s-2}$ , because  $|B'|$  is nonincreasing and  $y+x \leq \frac{3}{2}y \leq r_0$ . Also  $G(y) \geq G(x)$ . Hence

$$g(x) \geq \frac{2b_3(3/2)^{2s-2}}{1-2s} xG(x) \left( (2x)^{2s-1} - \left(\frac{2}{3}r_0\right)^{2s-1} \right),$$

and for  $x \leq x_0$  the bracket is at least  $\frac{1}{2}(2x)^{2s-1}$ .  $\square$

So the parity of the swirl decides the size of  $g$  at the stagnation point. For even swirl it is of order one. For odd swirl it is of order  $x^{2s}$ , in both directions: at least  $c_s x^{2s} G$ , and at most  $2G_1 \int_0^x B \leq Cx^{2s}$ .

Both local theories (Propositions 4.2 and 7.2) extend to (13). The source  $2gG_x$  is as smooth as  $G_x$ , because  $\partial_x^m g = B * \partial_x^m G$  and  $\partial_x G$  has compact support. The signs propagate as before:

- $G_x \geq 0$  on  $(0, \infty)$  is preserved by the continuity equation;
- $g \geq 0$  on  $(0, \infty)$  by Lemma 8.2;
- hence  $\omega \geq 0$  and  $u \leq 0$  on  $(0, \infty)$ .

In the identities (8) and Lemma 7.3 the buoyancy term becomes  $\int_\phi^\infty 2gG_y y^{\alpha-1} dy$ , since their derivation used only  $\omega_t = -u\omega_y + \text{source}$ .

**Theorem 8.3** (even swirl). *Let  $B$  be admissible with  $\int B = 1$ . Let  $G_0$  be even and smooth, with  $G_0(0) > 0$  and  $G'_0 \geq 0$  on  $(0, \infty)$  compactly supported. Let  $\omega_0$  be odd, smooth, compactly supported and nonnegative on  $(0, \infty)$ .*

- If  $0 < \alpha < 1$ , there are such data whose solution of (13), with either velocity law, is not global.*
- If  $\alpha \geq 1$  (local law) or  $1 \leq \alpha < 2$  (full law), every such solution is global.*

*Proof.* By Lemma 8.2(i),  $2G_{\min}G_x \leq 2gG_x \leq 2G_{\max}G_x$  on  $(0, \infty)$ , where  $G_{\min} = G_0(0)$  and  $G_{\max} = \sup G_0$  are conserved.

(a) The lower bound is the buoyancy  $\theta_x$  of  $\text{CKY}_\alpha$  and  $\text{HL}_\alpha$  with  $\theta = 2G_{\min}G$ , which is transported. The proofs of Theorems 5.1 and 7.6 use the buoyancy only through the lower bound  $\int_{\Phi_n}^{\Phi_{n-1}} \theta_y y^{\alpha-1} \geq \Phi_{n-1}^{\alpha-1}(\theta_{n-1} - \theta_n)$  and its positivity elsewhere. So they apply with  $G_0$  chosen so that  $2G_{\min}(G_{n-1} - G_n) = 2^{-n}$ .

(b) For the local law, repeat the proof of Theorem 6.1 with  $\eta = G_x$ . Step 1 is unchanged. In Step 2,  $\frac{d}{dt}(\omega\phi^\alpha) \leq 2G_{\max}(G_x\phi)\phi^{\alpha-1} \leq 2G_{\max}M_0R^{\alpha-1}$ , where  $R$  bounds the support. In Step 3 the buoyancy is at most  $2G_{\max}R^{\alpha-1}(G_{\max} - G_{\min})$ . For the full law, the proof of Theorem 7.7 needs only that same bound on the buoyancy.  $\square$

**Theorem 8.4** (odd swirl). *Let  $B$  be admissible. Let  $G_0$  be odd and smooth, with  $G_0 \geq 0$  and  $G'_0 \geq 0$  on  $(0, \infty)$ , and  $G'_0$  compactly supported. Let  $\omega_0$  be as in Theorem 8.3.*

- If  $0 < \alpha < \frac{1}{2}$ , there are such data whose solution of (13), with either velocity law, is not global.*
- If  $\alpha \geq \frac{1}{2}$ , every solution with the local law is global. For  $\alpha \geq 1$  we assume  $\int B = 1$ .*
- If  $1 \leq \alpha < 2$  and  $\int B = 1$ , every solution with the full law is global.*

*Proof.* (a) Choose the data with the support of  $G'_0$  inside  $(0, x_0]$ , and define  $x_n$  by  $G_0(x_n)^2 = \frac{1}{2}G_1^2 + 2^{-n}G_1^2$ . All the characteristics used stay in  $(0, x_0]$ . On  $[\Phi_n, \Phi_{n-1}]$ , Lemma 8.2(ii) gives

$$2gG_y y^{\alpha-1} \geq 2c_s G G_y y^{2\alpha-1} = c_s (G^2)_y y^{2\alpha-1} \geq c_s (G^2)_y \Phi_{n-1}^{2\alpha-1},$$

because  $2\alpha - 1 < 0$ . Since  $G^2$  is transported,

$$\int_{\Phi_n}^{\Phi_{n-1}} 2gG_y y^{\alpha-1} dy \geq c_s G_1^2 2^{-n} e^{(1-2\alpha)\psi_{n-1}}.$$

This is the forcing of the earlier proofs with  $1 - \alpha$  replaced by  $1 - 2\alpha$ . The Riccati step gives the recursion (10) with  $b = \frac{1-2\alpha}{2} > 0$ , and the induction is unchanged.

(b) For  $\frac{1}{2} \leq \alpha < 1$ , Lemma 8.2(ii) and  $B(r) \leq b_2 r^{2s-1}$  give  $g(x) \leq Cx^\alpha$  on  $[0, R]$ . Steps 1 and 4 of Theorem 6.1 are unchanged. Step 2 becomes  $\frac{d}{dt}(\omega\phi^\alpha) \leq 2g G_x \phi^\alpha \leq 2C(G_x\phi)\phi^{2\alpha-1} \leq$

$2CM_0R^{2\alpha-1}$ . In Step 3 the buoyancy is at most  $2C \int G_y y^{2\alpha-1} \leq 2CR^{2\alpha-1}G_1$ . For  $\alpha \geq 1$  use instead  $g \leq G_1$ , as in Theorem 8.3(b).

(c) As in Theorem 8.3(b).  $\square$

*Remark 8.5* (what is open). For the full law with odd swirl and  $\frac{1}{2} \leq \alpha < 1$ , that is  $\frac{1}{4} \leq s < \frac{1}{2}$ , we have neither blowup nor regularity. Two tools fail there:

- Step 1 of the local proof needs  $V_x \leq 0$ , which fails for the full law, because  $M_\alpha$  increases on  $(0, 1)$ ;
- the quadratic term of Lemma 7.3 has a sign only for  $\alpha \geq 1$ .

Scaling puts the threshold at  $\alpha = \frac{1}{2}$ . We expect regularity, but have no proof.

The results, with  $\alpha = 2s$ :

|            | local law                    | full law                                                           |
|------------|------------------------------|--------------------------------------------------------------------|
| even swirl | blowup iff $s < \frac{1}{2}$ | blowup for $s < \frac{1}{2}$ ; global for $\frac{1}{2} \leq s < 1$ |
| odd swirl  | blowup iff $s < \frac{1}{4}$ | blowup for $s < \frac{1}{4}$ ; global for $\frac{1}{4} \leq s < 1$ |

Here “blowup” means that some data of the class blow up, and “global” means that all data of the class are global.

## 9. NUMERICS

We solve (6) with particles, using  $\psi = -\ln \phi$  as the position variable and the adaptive integrator DOP853 (relative tolerance  $10^{-9}$ ). The labels are half log-spaced on  $[10^{-4}, 10^{-1}]$  and half uniform on  $[10^{-1}, 1]$ . The data are  $\omega_0 = 0$  and  $\rho_0 = \sin^2(\pi x/2)$ . We use  $N = 2^{12}$  and  $2^{13}$  particles (script `ckya.py`).

**Registered checks.** The following were written down before any analysis or run. Stop at  $t = 30$ , or when the innermost particle falls below  $x = 10^{-12}$ .

- N1: for  $\alpha \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ ,  $W = \Omega(t, x_{\min})$  grows by more than a factor 100 after it first reaches 1;  $1/W$  is closer to linear than to exponential in  $t$ ; and the two resolutions agree to 1%.
- N2: for  $\alpha \in \{\frac{5}{4}, \frac{3}{2}\}$ ,  $W$  grows by less than a factor 100.

Two deviations must be recorded.

- The registered stop rule refers to the particle *spacing* near 0 falling below  $10^{-12}$ . The code implements the *position* of the innermost particle falling below  $10^{-12}$ . Near 0 the spacing is about 0.003 times the position, so the implemented rule stops later. That favours N1, and N1 still fails below.
- One amendment was made after a low-resolution trial and before the registered runs. The innermost label was first  $10^{-8}$ , which made the stop rule fire after a compression of only  $e^9$ , even at  $\alpha = 0$ . We moved it to  $10^{-4}$  and left the criteria unchanged.

N1 fails at  $\alpha = \frac{1}{2}$  and  $\frac{3}{4}$ . The runs themselves are well resolved. What failed is that the stop rule, fixed in advance, allows a compression of about  $e^{18}$ . By (10),  $W$  grows like  $e^{(1-\alpha)\psi/2}$  at best, which is too slow near  $\alpha = 1$  to reach a factor 100 within that budget.

**Post hoc run (not registered).** To test that explanation we raised the stop rule to  $\psi = 300$ , with the same data and method (`ckya_deep.py`). At  $\alpha = 0.5, 0.75$  and  $0.9$ ,  $W$  grows by factors of 560, 620 and 2800. In each case  $1/W$  is linear in  $t$  with relative residual below  $10^{-4}$ , against  $3\text{--}5 \times 10^{-2}$  for  $\ln W$ . The linear fit extrapolates to a blowup time just past the end of each run: 3.29, 6.86 and 18.19, against run ends of 3.28, 6.84 and 18.18. At  $\alpha = 0.5$  the run ended when the integrator’s step size collapsed at  $t = 3.28$ , before the stop rule. The two resolutions agree

| $\alpha$ | growth of $W$         | $1/W$ nearer linear | resolutions agree  | verdict  |
|----------|-----------------------|---------------------|--------------------|----------|
| 0        | 525                   | yes                 | $1 \times 10^{-5}$ | N1 holds |
| 0.25     | 222                   | yes                 | $1 \times 10^{-6}$ | N1 holds |
| 0.5      | 48                    | yes                 | $7 \times 10^{-7}$ | N1 fails |
| 0.75     | 8.7                   | no                  | $3 \times 10^{-7}$ | N1 fails |
| 1.25     | peak 1.4, then decays | –                   | $2 \times 10^{-4}$ | N2 holds |
| 1.5      | peak 1.1, then decays | –                   | $3 \times 10^{-4}$ | N2 holds |

TABLE 1. Registered checks, applied mechanically.

to  $4 \times 10^{-4}$  or better. The measured  $d \ln W / d\psi$  is about two thirds of  $(1 - \alpha)/2$  at all three values.

**The ingredients.** On the same runs (`verify_thm.py`) we checked four things. The identity (8) holds to  $5 \times 10^{-4}$ , which is the accuracy of the finite differences in time.  $\eta x$  never increases along any characteristic, at any  $\alpha$ . For  $\alpha \in \{1, \frac{5}{4}, \frac{3}{2}\}$  up to  $t = 20$ , the bound of Step 2 holds with equality at worst, and  $W$  stays below the bound of Step 3 by at least 0.5.

**The full law.** For  $HL_\alpha$  we use the same particles. The velocity (12) is summed directly over particle positions with trapezoid weights, using  $\ln(x_i/x_j)$  to avoid underflow. The integrator is RK45 with relative tolerance  $10^{-8}$  (`hla.py`). The data are  $\omega_0 = 0$  and  $\theta_0 = \sin^2(\pi x/2)$  on  $[0, 1]$ , extended evenly and by 1 beyond. We use  $N = 512$  and 1024.

After the analysis of Section 7, and before any simulation of  $HL_\alpha$ , we registered the following. Stop when  $\psi > 300$ , when the step size collapses, or at  $t = 60$ .

- N3: for  $\alpha \in \{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ , the same three criteria as N1.
- N4: for  $\alpha \in \{\frac{5}{4}, \frac{3}{2}\}$ ,  $A(t, 0) \leq A(0, 0) + t$ , which is the bound in the proof of Theorem 7.7, and  $W$  grows by less than a factor 100.
- N5: at probe particles, (i)  $V \geq 2\alpha A$ ; (ii) the cross term is nonnegative; (iii) the self term is at least  $-\alpha^2 A^2$  for  $\alpha < 1$  and at most 0 for  $\alpha \geq 1$ ; and (iv) Lemma 7.3 holds to relative accuracy  $10^{-2}$ .

Scripts `hla_an.py` and `hla_posthoc.py` apply these criteria.

| $\alpha$ | end of run                      | growth since $W = 1$  | $1/W$ linear   | resolutions          |
|----------|---------------------------------|-----------------------|----------------|----------------------|
| 0.25     | step collapse, $t \approx 2.70$ | 62, 60                | yes            | $4 \times 10^{-2}$   |
| 0.5      | step collapse, $t \approx 3.20$ | 650, 580              | too few points | $1.2 \times 10^{-1}$ |
| 0.75     | step collapse, $t \approx 5.70$ | 210, 195              | yes            | $1.0 \times 10^{-1}$ |
| 1.25     | horizon $t = 60$                | peak 1.4, then decays | –              | $4 \times 10^{-3}$   |
| 1.5      | horizon $t = 60$                | peak 1.1, then decays | –              | $6 \times 10^{-4}$   |

TABLE 2. Registered checks for the full law. Growth is given for  $N = 512$  and  $N = 1024$ .

Applied mechanically, the verdicts are as follows.

- N3 fails at every  $\alpha$  on the resolution criterion. It also fails at  $\alpha = \frac{1}{4}$  on growth, because the step size collapses at  $60\times$ . At  $\alpha = \frac{1}{2}$  the linearity criterion cannot be evaluated, because the last decade contains only two outputs.
- N4 holds.

- N5 (i)–(iii) hold at every output and probe for  $\alpha < 1$ . For  $\alpha \geq 1$ , (i) and (ii) fail by up to  $1.2 \times 10^{-5}$  and  $5.5 \times 10^{-2}$ . Lemma 7.4 makes no claim there: for  $\alpha > 1$ ,  $M_\alpha < 2\alpha$  near  $s = 1$  and  $P > 0$  somewhere on  $(1, \infty)$ . That part of the registration was too broad.
- N5 (iv) holds at  $\alpha = \frac{1}{2}, \frac{3}{4}$  and fails at  $\alpha = \frac{1}{4}$  ( $1.3 \times 10^{-2}$ ),  $\frac{5}{4}$  (0.18) and  $\frac{3}{2}$  ( $2 \times 10^{-2}$ ).

*Post hoc diagnosis (not registered).*

- The fitted blowup times of the two resolutions agree to  $6 \times 10^{-4}$  or better, and  $W$  agrees to  $4 \times 10^{-3}$  for  $t \leq 0.9T$ . The registered failures come from the approach to  $T$ : there  $W \sim (T - t)^{-1}$ , so a shift of 0.05% in  $T$  becomes 10% in  $W$ .
- At  $\alpha = \frac{1}{2}$  with dense output,  $W$  grows by a factor  $2.4 \times 10^4$ , and  $1/W$  is linear with residual  $6 \times 10^{-6}$ , against  $3 \times 10^{-2}$  for  $\ln W$ . The fit extrapolates to  $T = 3.2072$ .
- For  $\alpha \geq 1$  the worst identity residuals sit at the outermost particle, where every term is of size  $10^{-8}$  and the terms cancel. Measured against the largest term, the residual is below  $6 \times 10^{-3}$ .
- At  $\alpha = \frac{1}{4}$  the residual came from differencing in time over outputs spaced 0.1 apart. With spacing  $10^{-3}$  it is  $1.1 \times 10^{-3}$  at  $N = 512$  and  $4.7 \times 10^{-4}$  at  $N = 1024$ .

## 10. DISCUSSION

**What the dichotomy says.** In both wall models, the local  $\text{CKY}_\alpha$  and the full  $\text{HL}_\alpha$ , the  $H^s$  family blows up exactly below  $s = \frac{1}{2}$ . This is the off-axis threshold that scaling gives in [14]. Both models are regular at and above it. The method does not lose anything at the threshold: the growth rate  $e^{(1-\alpha)\psi/2}$  in (10) vanishes exactly there, for either kernel.

**A registered expectation that failed.** Before attempting the full law we registered the expectation, with confidence 55%, that the argument of [6] would extend only to  $\alpha < \alpha_0$  for some  $\alpha_0 < 1$  strictly. That was wrong. Their functional argument does not extend at all (Remark after Theorem 7.7). What extends is their positivity idea, and only in the weakened form of Corollary 7.5, where a lower bound  $-c_\alpha A^2$  replaces exact positivity. Inside the characteristic argument that lower bound is enough, all the way to  $\alpha = 1$ .

**The parity of the swirl.** Section 8 shows that the derived buoyancy  $2gG_x$  sees the parity of the swirl, which  $\theta_x$  does not.

- For even swirl,  $g$  is of order one at the stagnation point. The threshold stays at  $s = \frac{1}{2}$  (Theorem 8.3).
- For odd swirl,  $g$  is of order  $x^{2s}$  (Lemma 8.2). The buoyancy is suppressed, and the threshold moves to  $s = \frac{1}{4}$ . This is proved for the local law, and proved on the blowup side for the full law (Theorem 8.4).

A direct computation (`parity_g.py`, `parity_lemma.py`) shows the same thing on compressed fronts, for  $s \in \{0.1, 0.2, 0.25, 0.35, 0.4, 0.45\}$  and widths down to  $10^{-3}$ . At  $s = 0$  parity is invisible, since only  $G^2$  enters.

The initial data of Luo and Hou [12], and Hou’s interior datum used in [14], have odd swirl. So in the parity of the known blowup candidates the window for the  $H^s$  family is  $0 < s < \frac{1}{4}$ , not  $0 < s < \frac{1}{2}$ . This corrects the Boussinesq count of [14] away from the axis, which assumed that  $g$  is carried by its large-scale part. The Jacobian count there uses only the small-scale part of  $g$ , and stays at  $s < \frac{1}{2}$ . The early-time reversal between  $s = \frac{1}{4}$  and  $s = \frac{1}{2}$  observed in [14] is compatible with both.

**What it does not say.** With the derived buoyancy, one reduction separates the model (13) from (1) on a cylinder: the column ansatz of Hou and Luo, which also removes the wall term  $G_y g_x$  of Lemma 2.2. For even swirl there is one further choice, an infrared scale in the kernel, which stands in for the spectral gap of the cylinder. At  $s = 0$ , removing the local-part reduction took [6, 4], and removing the column ansatz took [1, 2].

None of this bears on Question 7.1 on  $\mathbb{R}^3$  or  $\mathbb{T}^3$ , where there is no wall [14]. The results are statements about the  $H^s$  family on a domain with boundary.

**The boundary condition.** Remark 2.1 shows that the choice of boundary condition for the inertia operator decides whether the wall scenario exists at all. For the  $H^s$  family on a domain with boundary this choice is part of the definition of the metric. The question of [14] on  $\mathbb{R}^3$  and  $\mathbb{T}^3$ , where there is no boundary, is not affected by anything here.

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The scripts underlying the computational statements are available from the author. They are `derive_wall.py`, `ckya.py`, `ckya_deep.py`, `verify_thm.py`, `kernel_hl.py`, `hla.py`, `hla_an.py`, `hla_posthoc.py`, `parity_g.py` and `parity_lemma.py`.

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